

The Role of the Corticospinal Tract in the Evolution of Human Digital Dexterity

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Abstract. A morphometric analysis of the corticospinal tract's relation to digital dexterity was performed on 21 species theoretically related to man's ancestral lineage. The results indicate that the Primate line is not unique among mammals with respect to the cortical control of digital dexterity. A comparative analysis suggests that two changes took place early in Primate evolution: a reduction in functional distance (i.e. number of synapses) between neocortex and spinal motor neurons innervating the digits, and an extension of direct neocortical influence beyond the cervical segments of the spinal cord. A further change progressed throughout Primate evolution, from the mid-Eocene to the present, in which the overall size of the corticospinal tract increased steadily as though consolidating the cortical influence over body musculature, especially that of the digits.

It has been suggested in the past that the corticospinal tract has evolved in Primates in close association with the evolution of fine control of the hand and digits [e.g. Phillips, 1971; Phillips and Porter, 1977]. This suggestion is founded on the results of anatomical, electrophysiological, and ablation-behavior experiments. For example, electrophysiological evidence for direct synapses of corticospinal fibers onto the motor neurons of the hand and forearm are virtually restricted to Primates [Felix and Wiesendanger, 1971; Landgren et

al., 1962; Phillips and Porter, 1964]; although a revealing exception is the raccoon which also possesses sparse corticomotoneuronal connections and is also known for its manipulative skills [Wirth et al., 1974]. The electrophysiological findings are closely paralleled by descriptions of the anatomical connections of the corticospinal fibers in the spinal cord [Heffner and Masterton, 1975]. Moreover, in ablation-behavior studies of impaired function after the corticospinal fibers have been severed, Primates have consistently de-

monstrated deficits in fine manipulative movements of the digits. Where special efforts in retraining have not been provided, these deficits have been permanent [Lawrence and Kuypers 1968; Tower, 1940; Woolsey et al., 1972]. When specifically retrained, some recovery of discrete finger movements can occur, though these never appear entirely normal [Bucy et al., 1964; Chapman and Wiesendanger, 1982; Hepp-Reymond and Wiesendanger, 1972; Schwartzmann, 1978].

Thus, the corticospinal motor system seems to make an important contribution to digital dexterity in Primates and may have undergone evolutionary changes in parallel with the evolution of digital dexterity. In this report we attempt to reconstruct the evolutionary histories of the corticospinal tract and digital dexterity by tracing these features across a sequence of extant mammals representing successive grades of neurological similarity to mankind. This approach in no way denies the importance of the many other fiber systems, both sensory and motor, which make possible the fine independent digital movements of which some animals are capable.

In a previous report, we made a similar morphometric analysis of the relation between the variation in the corticospinal tract and the variation in digital dexterity in 69 unselected mammalian species [Heffner and Masterton, 1975]. In this report we narrow the analysis to a subset of mammals whose nervous systems are thought to most closely approximate the nervous systems of the successive grades of extinct animals suggested as making up mankind's ancestral lineage [cf. Gould, 1976]. Perhaps, needless to say, this goal requires

several judgments regarding which of the available animals are most suitable for providing data to approximate the condition of the extinct ancestral animals that would be the ideal subjects of this study. Although these judgments are always arguable and never completely satisfying, reasonably good data regarding the morphology of the corticospinal tract in extant mammals has now accumulated to the point that an attempt to trace its phylogeny may be justified. Nevertheless, the sequence of morphological and behavioral changes which are the subject of this report are inferred entirely from the differences between grades of extant species. Therefore, the evolutionary conclusions must be weakened accordingly.

Materials and Methods

Sample Selection and the Assignment of Phyletic Levels

To narrow the question to man's ancestral lineage, we have selected from our original population of data on 69 mammalian species [Heffner and Masterton, 1975] a subset of 21 species based solely on their relative lack of neural and manual specialization as compared to other mammals with the same propinquity with mankind (fig. 1). That is, the species included in the present analysis are those whose nervous systems and forepaws are most likely to resemble those of the inferred common ancestry of a much larger group (i.e., see fig. 1, branchpoints 1 through 8). The necessity of selecting the least neurologically specialized species required the elimination of most marsupials (e.g. kangaroos, phalangers; branchpoint 1, fig. 1) and all placental mammals other than Insectivora and Primates (branchpoint 2, fig. 1). Likewise, the criterion of selecting the least manually specialized species (while maintaining as many species as possible in the sample) required the elimination of several otherwise good candidates. For example, marmosets were eliminated due to the likelihood that their hands are degenerate rather than primitive.

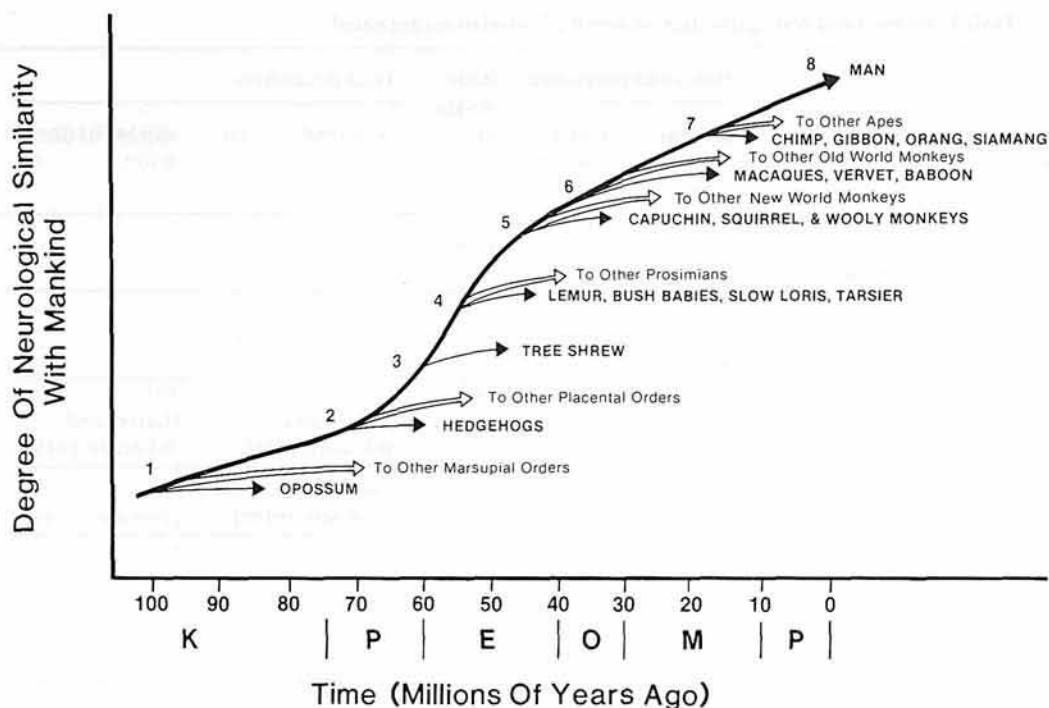


Fig. 1. Phylogenetic relationship between some living mammals and inferred mammals in man's ancestral lineage. The numbers 1 through 8 representing successive branchpoints and thus relative propinquity, indicate degree of neurological similarity with mankind and are used in the correlations to represent phyletic level.

Spider monkeys were also eliminated due to their degenerate thumbs [Napier and Napier, 1967]. Finally, hedgehogs were retained in the sample (even though they have lost the more dexterous convergence of digits possessed by some Marsupials and some other Insectivora) because they are the least specialized of the Insectivora on which data are available.

These successive selections of the most appropriate representatives culminated in the approximation of a pre-Eutherian grade with one species (Virginia opossums), a ground-dwelling Insectivore grade with two species (European and Pakistani hedgehogs), a tree-dwelling Insectivore grade with one species (tree shrew), a Prosimian grade with four species (large-tailed and Senegalese bushbabies, ring-tailed lemur and slow loris), a new world Simian grade with three species (squirrel, wooley and Cebus monkeys), an old world Simian grade with three species (two ma-

caques and a baboon), a Pongid grade with four species (gibbon, chimp, orang, and siamang), and finally, the Hominid grade with one species. Formal names and the relevant data for each species are given in table I.

Derivation of Numerical Parameters

Digital Dexterity Scale. A method for ranking digital dexterity in a wide range of mammals has been described previously [Heffner and Masterton, 1975]. It is based primarily on Napier's analysis of hand anatomy and the hand's normal use [Napier and Napier, 1967]. A rank (from 1 to 7) has been assigned according to an animal's relative ability to make skilled and independent movements of its digits. The meanings of the ranks are given in table II. It should be noted that anatomical features of the hand are an important consideration in the ranking despite the fact

Table I. Animals and descriptive data on which all correlations are based

Animal	Behaviour parameter		Body weight kg	Tract parameters	
	phyletic level	digital dexterity		area, mm ²	number of fibers × 10 ⁻³
Bushbaby (<i>Galago crassicaudatus</i>)	4	5	1.14		
Bushbaby (<i>Galago senegalensis</i>)	4	5	0.26		
Chimpanzee (<i>Pan troglodytes</i>)	7	6	44.0	7.77 [Lassek and Wheatley, 1945]	807 [Lassek and Wheatley, 1945]
Gibbon (<i>Hylobates lar</i>)	7	6	6.0	3.44 [Verhaart, 1970b]	315 [Verhaart, 1948b]
Hedgehog (<i>Erinaceus</i>)	2	2	0.75	0.02 [Verhaart, 1970b]	
Hedgehog (<i>Paraechinus</i>)	2	2	0.5		
Lemur (<i>Lemur catta</i>)	4	5	2.1		
Man (<i>Homo sapiens</i>)	8	7	70.0	11.43 [Lassek and Rasmussen, 1940]	1,101 [Lassek and Rasmussen, 1940]
Monkey (<i>Cebus apella</i>)	5	5	2.4	2.0 [Manocha et al., 1968]	
Monkey (<i>Cercopithecus pygerythrus</i>)	6	6	4.0		
Monkey (<i>Lagothrix lagotricha</i>)	5	5	6.0		
Monkey (<i>Macaca ira</i>)	6	6	5.0	1.18 [Verhaart, 1954]	195 [Verhaart, 1948a]
Monkey (<i>Macaca mulatta</i>)	6	6	8.5	2.89 [Lassek, 1941]	400 [DeMeyer and Russel, 1958]
Monkey (<i>Papio</i>)	6	6	25		
Monkey (<i>Saimiri sciureus</i>)	5	5	0.75	0.70 [Verhaart, 1970b]	
Opossum (<i>Didelphis</i>)	1	3	3.7	0.22 [Towe, 1973]	48.1 [Towe, 1973]

average fiber size μm	largest fiber size μm	Termination parameters		
		extension down spinal cord (rank)	spinal lamina receiving synapses	
			ventralmost	densest
		8 [Ferrer, 1971]	9 [Ferrer, 1971]	6 [Ferrer, 1971]
	2.5 [Verhaart, 1966]	9 [Goode and Haines, 1975]	9 [Goode and Haines, 1975]	6 [Goode and Haines, 1975]
		9 [Petras, 1968]	10 [Petras, 1968]	6 [Petras, 1968]
1.09	12 [Verhaart, 1970b]	8 [Petras, 1968]	10 [Petras, 1969]	6 [Petras, 1968]
		1 [Verhaart, 1970b]		
		1 [RoBards, pers. commun.]		
	3 [Verhaart, 1966]	7 [Verhaart, 1966]		
1.03	22 [Truex and Carpenter, 1969]	9 [Truex and Carpenter, 1969]		
		9 [Petras, 1968]	10 [Petras, 1969]	6 [Petras, 1968]
		9 [Petras, 1968]	10 [Petras, 1969]	6 [Petras, 1968]
		9 [Petras, 1968]	10 [Petras, 1968]	6 [Petras, 1968]
0.60				
0.72		9 [Petras, 1968]	10 [Petras, 1969]	6 [Petras, 1968]
			10 [Phillips, 1967]	
	8 [Verhaart, 1970b]	8 [Verhaart, 1966]	9 [Harting and Noback, 1970]	6 [Harting and Noback, 1970]
0.45		3 [Bautista and Matzke, 1965]	8 [Martin and Fisher, 1968]	5 [Martin and Fisher, 1968]

Table I (continued)

Animal	Behaviour parameter		Body weight kg	Tract parameters	
	phyletic level	digital dexterity		area, mm ²	number of fibers × 10 ⁻³
Orangutan (<i>Pongo pygmaeus</i>)	7	6	60.0	2.61 [Verhaart, 1970b]	
Siamang (<i>Hylobates syndactylus</i>)	7	6	10.0	3.25 [Verhaart, 1948b]	239.7 [Verhaart, 1948b]
Slow loris (<i>Nycticebus coucang</i>)	4	5	1.3	0.28 [Verhaart, 1966]	
Tarsier	4	4	0.12	0.16 [Tilney, 1927]	
Tree shrew (<i>Tupaia glis</i>)	3	3	0.16	0.12 [Tigges and Shantha, 1969]	

Table II. Scale of digital dexterity

	Rank	Description
Specialization for locomotion	1	↑ fused or restrained digits separate digits that do not converge when flexed
	2	
Most primitive form	3	convergent but not prehensile (not capable of holding an object in one hand)
Specialization for manipulation	4	↓ prehensile digits, non-opposable thumb prehensile digits, pseudo-opposable thumb opposable thumb, capable of power grip and limited precision opposable thumb, capable of precision grip in opposition to each finger
	5	
	6	
	7	

Scale derived from Bishop [1964], Napier [1960, 1961], and Napier and Napier [1967].

that digital dexterity is a behavioral rather than a physical character. However, the ranking scheme used here is in close agreement with the few behavioral studies of dexterity presently available [e.g. Welles, 1976].

Corticospinal Tract Morphology. This report focuses on three morphological features of the corticospinal tract because they were the ones found to be related to digital dexterity among the larger sample of mammals [Heffner and Masterton, 1975]. Briefly they are: the area of the corticospinal tract (an indication of its overall size), the distance it extends down the spinal cord (an indication of its range of influence over the entire body), and the ventralmost lamina of the spinal cord gray matter receiving corticospinal terminals (an indication of the directness of its influence on the activity of spinal motor neurons). Other features, possibly equally plausible, are discussed only briefly because in our previous analysis they were found to be unrelated to digital dexterity [Heffner and Masterton, 1975].

Size of the Corticospinal Tract. The size of the corticospinal tract is used as a measure of its overall development [Lassek, 1954; Verhaart, 1970a; Heffner and Masterton, 1975]. Briefly, the cross sectional area of the tract (in mm²) as it passes through one pyramid at the level of the lower medulla is obtained from suitable histological sections. The logarithm of this

average fiber size μm	largest fiber size μm	Termination parameters		
		extension down spinal cord (rank)	spinal lamina receiving synapses	
			ventralmost	densest
1.35				
	4.5 [Verhaart, 1970a]	8 [Campbell, 1965]	10 [Campbell et al., 1966]	6 [Campbell et al., 1966]
	2 [Verhaart, 1966]	5 [Shriver and Noback, 1967]	6 [Jane et al., 1969]	5 [Campbell, 1965]

value is used for calculations. Measurement error is introduced by variation in the exact level of the medulla from which the available measure was taken and by the existence of many fibers in the pyramids that do not reach the spinal cord, synapsing instead in the lower medulla, and are thus not true corticospinal fibers.

Extension of the Tract Down the Spinal Cord. The most caudal spinal cord segment to which the tract extends is closely related to mammalian dexterity though it is not yet clear why this is so [Lassek, 1954; Heffner and Masterton, 1975]. For present purposes, the extension of the tract down the spinal cord is assigned a rank based on the most caudal segment of the cord known to receive corticospinal terminations and simply indicates the rostral-caudal extent of the spinal cord receiving direct cortical influence. The ranking is arbitrary but systematic, with 1 indicating upper cervical levels and 9 indicating the coccygeal level. Measurement error is introduced in this scale by the inclusion of data obtained with histological techniques of varying power. For example, in the bushbaby a normal fiber stain showed axons penetrating only as far as the lumbar level [Verhaart, 1966] but the use of a Fink-Heimer preparation revealed axons extending to coccygeal levels [Goode and Haines, 1975]. Use of autoradiographic techniques similarly improves the ability to visualize

these pathways [cf. Harting and Noback, 1970, and Tigges et al., 1979, on the squirrel monkey]. Where available, data from degeneration studies are used rather than data obtained from normal material. However, due to the relative newness of the autoradiographic technique, such data are as yet available only for one species (squirrel monkey) and are not included here.

Ventralmost Lamina of Spinal Cord Gray Matter Receiving Direct Corticospinal Synapses. This measure is an indirect indication of what functional types of spinal cord elements (sensory neurons, internuncial neurons, distal dendrites or somata of motor neurons) receive direct cortical input via the corticospinal tract. The rank assigned is the ventralmost lamina (using Rexed's [1954] scheme) in which corticospinal axon terminals have been found. Lower ranks (1 to 4) indicate synapses only on neurons in the more dorsal sensory regions, middle ranks (5 to 8) indicate synapses in the central internuncial regions (which include distal dendrites of motor neurons), a rank of 9 indicates synapses in the region of motor neurons and their proximal dendrites in lamina IX, and a rank of 10 indicates that synapses have been observed on the cell bodies of motor neurons themselves. As can be seen in table 1, all mammals studied thus far have corticospinal synapses on sensory and internuncial elements, but relatively few

have synapses on the proximal dendrites (rank 9) or somata (rank 10) of spinal motor neurons.

Other Corticospinal Tract Parameters. Although their values are also listed in table I, four other parameters of the corticospinal tract will be considered only briefly because each alone was found to be unrelated to digital dexterity among a larger sample of mammals. These are: the lamina of spinal cord gray matter receiving the most corticospinal terminals, the number of fibers in the corticospinal tract, the average size of the fibers, and the size of the largest fibers.

Body Size. It has been shown previously that large mammals have many large muscles and large motor tracts to serve them. Consequently, in any analysis of motor systems which includes animals whose body sizes differ greatly, gross body size is an important and potentially confounding factor: the size of motor structures relative to body size is usually a more important factor than their absolute size [Jerison, 1973; Towe, 1973]. The measure of body size employed here is simple body weight in kilograms. The specific values given in table I are the midranges taken from Walker [1968] or Napier [1967]. Logarithms of these values were used in the statistical calculations.

Results

It is convenient to present the results in three sections. The first deals with data pertinent only to considering the evolution of mankind's digital dexterity. The second section deals with data pertinent only to considering the evolution of mankind's corticospinal tract. The final section addresses the relationship between the two.

Evolution of Human Digital Dexterity

The idea that a phyletic sequence of extant mammals might reflect a plausible sequence of increments in digital dexterity is supported by the data shown in figure 2. This figure illustrates the differences in digital dexterity from the primitive conver-

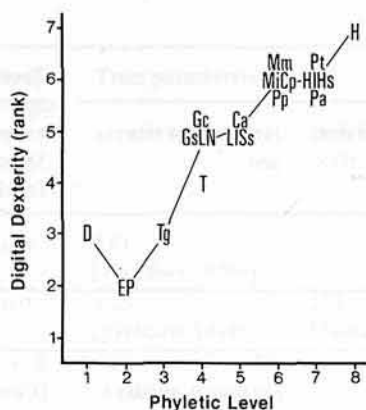


Fig. 2. Relation between digital dexterity and phyletic level for mammals in sample. Although each species was selected for its relative lack of neurological and manual specialization as compared to other mammals with the same propinquity with mankind, there appears to be a steady increase in digital dexterity with the only reversal at the level represented by hedgehogs, neurologically primitive animals which have developed forepaws somewhat specialized for digging. Ca, *Cebus apella*; Cp, *Cercopithecus pygerythrus*; D, *Didelphis*; E, *Erinaceus*; Gc, *Galago crassicaudatus*; Gs, *Galago senegalensis*; H, Human; HI, *Hylobates lar*; Hs, *Hylobates syndactylus*; L, *Lemur*; LI, *Lagothrix lagotricha*; Mi, *Macaca ira*; Mm, *Macaca mulatta*; N, *Nycticebus coucang*; P, *Paraechinus*; Pa, *Papio*; Pp, *Pongo pygmaeus*; Pt, *Pan troglodytes*; Ss, *Saimiri sciureus*; T, *Tarsier*; Tg, *Tupaia glis*.

gent digits of the opossum to the precision grip of man. There is only one reversal in an otherwise ascending curve: at the 'primitive Eutherian' level represented here only by hedgehogs. (It should be noted that hedgehogs, despite their neurological primitiveness, have forepaws somewhat specialized for digging.)

Although the correlation between digital dexterity and phyletic level illustrated in figure 2 is very high and reliable ($r = 0.92$, $p < 0.01$), it is complicated by the close relation of each of those two parameters to

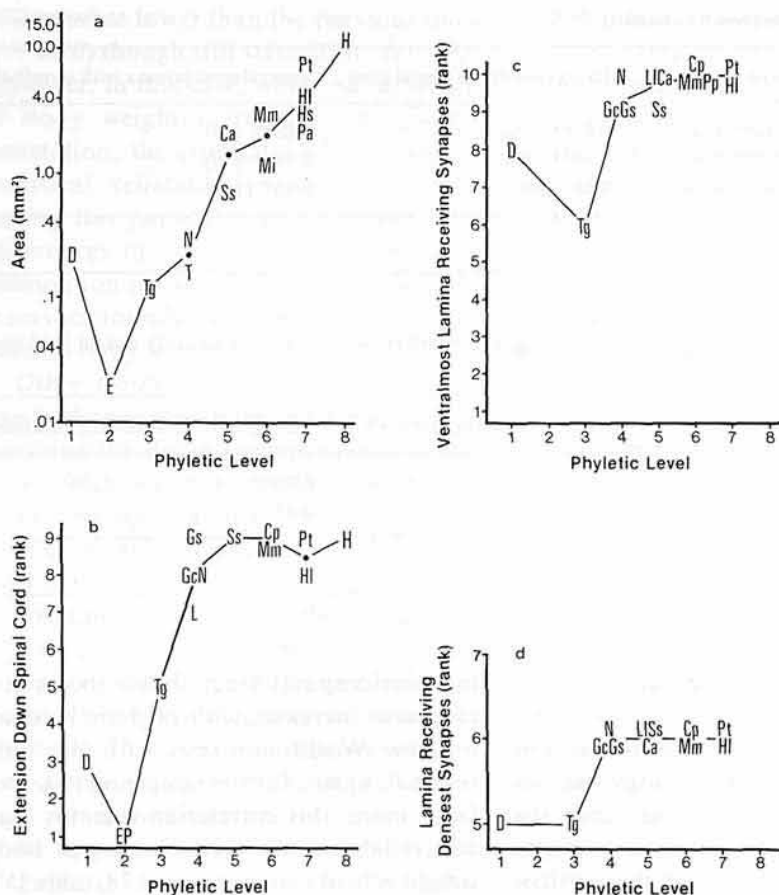


Fig. 3. Relation between phyletic level and four morphological features of the corticospinal tract. Note that each feature shows systematic change with phyletic level and that area, extension down the spinal cord, and ventralmost spinal lamina with terminals each show a reversal at a primitive eutherian grade (2) similar to the reversal seen in figure 2. For abbreviations see figure 2.

body size. However, by using partial correlations in which one variable (e.g. body weight) can be held mathematically constant in the correlation between two other variables, it can be shown that the correspondence between digital dexterity and phyletic level is not due to their common correlation with body weight. That is, the partial correlation between digital dex-

terity and phyletic level, with body weight held constant, is 0.84, whereas the correlation between digital dexterity and body weight with phyletic level held constant is only 0.10. Thus, it is reasonably safe to conclude that the increase in digital dexterity with phyletic level is *not* merely a secondary result of the concomitant increase in body size.

Table III. Correlations (row by column)

	Area	Extension down spinal cord	Ventralmost lamina with terminals
Digital dexterity	0.94** (14)	0.92** (16)	0.86** (13)
Phyletic level	0.89** (14)	0.80** (16)	0.70** (13)
Body weight	0.81** (14)	0.44 (16)	0.66* (13)

* $p < 0.05$; ** $p < 0.01$.**Table IV.** Partial correlations holding body weight constant (row by column with body weight held constant)

	Area	Extension down spinal cord	Ventralmost lamina with terminals
Digital dexterity	0.89**	0.95**	0.75**
Phyletic level	0.74**	0.78**	0.42

** $p < 0.01$.

Evolution of the Corticospinal Tract

Area. Turning now to the differences in corticospinal tract morphology across phyletic levels, figure 3a illustrates the close correspondence between phyletic level and the overall size of the corticospinal tract. The correlation between the two is 0.89 (table III) with the only reversal, again, at the level represented by hedgehogs. Once more, the increase in tract size with the higher phyletic levels is not due to the parallel increase in body weight as shown by a partial correlation coefficient (with weight held constant) of 0.74 (table IV). Thus, the increase in cross-sectional size of the corticospinal tract with advances in phyletic level is beyond that which would be expected from parallel increases in body size alone.

Extension of the Corticospinal Tract Down the Spinal Cord. The extension of

the corticospinal tract down the spinal cord also increases with phyletic level up to New World monkeys with the only reversal, again, for the hedgehog (fig. 3b). Once more, this correlation remains high and reliable when the influence of body weight is held constant ($r = 0.78$, table IV). Thus, like tract size, the increase in the tract's extent with the increase in phyletic level is also far beyond that which would be expected from the parallel increase in body size alone.

Ventralmost Lamina of Termination. The relation between phyletic level and the ventralmost lamina of spinal cord gray receiving corticospinal terminals is illustrated in figure 3c. In contrast to the previous relationships, this one has a reversal at the grade represented by tree shrews. (Data for this parameter are not available for hedgehogs.) Perhaps because of the lack of data on primitive grades, this correlation

is somewhat lower than the previous ones ($r = 0.70$) though still statistically reliable. However, in this case, when the influence of body weight is removed by partial correlation, the correlation falls short of statistical reliability ($r = 0.42$). Thus, among this particular set of species, the differences in the ventralmost lamina of termination across phyletic levels still remains confounded with the parallel differences in body weight.

Other Corticospinal Tract Parameters. The lamina of the spinal cord gray matter receiving the densest corticospinal terminals, which was not correlated with dexterity among mammals in general, is correlated among the restricted species considered here ($r = 0.89$). The correlation between phyletic level and the spinal cord lamina receiving the densest projection (fig. 3d) is reliable ($r = 0.75$) and is not reduced when body weight is held constant ($r = 0.78$). However, because this parameter assumes only two values in the limited data presently available, the significance of the apparent shift in the modal region of corticospinal synapses cannot be evaluated with confidence.

Phyletic level is *not* reliably correlated with the size or the number of the fibers in the corticospinal tract in the sample considered here. Furthermore, both correlations fall even lower when the contribution of body weight is eliminated. However, due to the demonstrated difficulty of counting the many small unmyelinated fibers which make up the majority of the corticospinal tract even in man, we cannot rule out that these factors may yet prove to be important when counts made with the electron microscope become available [Leenen et al., 1982].

Relationship between the Evolution of Digital Dexterity and the Evolution of the Corticospinal Tract

So far we have seen that digital dexterity, the size of the corticospinal tract, the extension down the spinal cord of fibers of cortical origin, and possibly the directness of their influence on spinal motor neurons are each, alone, highly and reliably correlated with phyletic level. The relation between digital dexterity and each of the morphological parameters remains to be seen.

The correlations between digital dexterity and each of the morphological variables are given in table III and are illustrated in figure 4. They range from 0.86 to 0.94 and are each highly reliable. Once again, these direct correspondences between corticospinal characteristics and digital dexterity are not significantly affected by the statistical removal of the influence of body size (table IV). In this respect, the subsample of extant species chosen here to approximate mankind's ancestral lineage is *not* appreciably different from mammals in general [cf. Heffner and Masterton, 1975].

However, in seeking the correspondence between changes in corticospinal tract morphology and changes in digital dexterity that might have occurred in mankind's lineage, certain further constraints are imposed by the timing of the changes themselves. That is, in order for a change in some aspect of the corticospinal tract to be considered as a possible mechanism for an increment in digital dexterity, the morphological change must occur either before or at the same stage as the change in dexterity. A comparison of figure 2 with figure 3 provides further insight into the

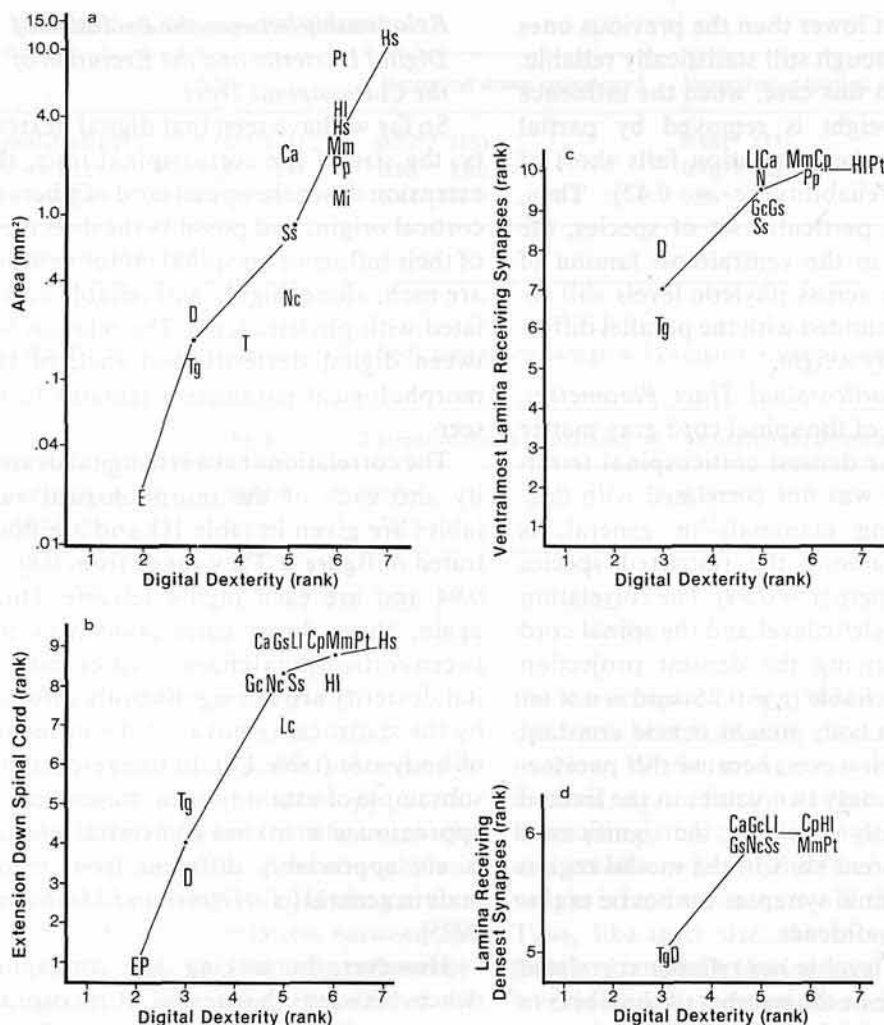


Fig. 4. Relation between digital dexterity and four morphological features of the corticospinal tract. Note similarity to corresponding parts of figure 3. For abbreviations see figure 2.

possible successive contributions of the corticospinal tract to digital dexterity.

Influence of Increases in Tract Size. The size of the corticospinal tract corresponds approximately to the number and size of the fibers of which it is composed though neither number nor fiber size alone is presently correlated reliably with dexterity

(possibly due to inaccurate counts and unavailable data for many species). Nevertheless, the size of the pyramid in the lower medulla can be loosely interpreted as an indication of the overall amount of influence of the corticospinal system on the body's musculature. As illustrated in figure 5, changes in the area of the tract with

higher phyletic grades parallel the changes in digital dexterity very closely, both even reversing together at the 'primitive Eutherian' level. Further, their correlation ($r = 0.94$) (fig. 4a) remains relatively unchanged ($r = 0.89$) with the removal of body weight as a factor. Still further, and unlike the other parameters which are related to the mode of termination of the tract in the spinal cord, the size of the tract continues to increase up to the highest phyletic levels. Thus, this indirect measure of the overall amount of influence of the cerebral cortex on the spinal cord is the parameter most closely related to the progressive changes in digital dexterity across phyletic levels.

Influence of Changes in Corticospinal Terminations. The two morphological parameters representing the extent of the body musculature influenced by the corticospinal tract and the directness of that influence (i.e. 'extension down spinal cord' and 'ventralmost lamina with corticospinal terminals') appear to take similar evolutionary courses. As figures 3b and 3c show, both parameters reach a maximum at the Anthropoid Primate level and undergo no further change among higher grades. Thus, either or both of these features may have underlain early advances in digital dexterity (and indeed may have done so as well in other mammalian Orders such as Carnivores in which the corticospinal tract also extends the length of the spinal cord in some species [Nyberg-Hansen and Brodal, 1963]). The contribution of the shift in the region of densest corticospinal terminals to digital dexterity (fig. 3d, 4d) is even more difficult to interpret. Though the shift of influence closer to motor neurons does seem to be main-

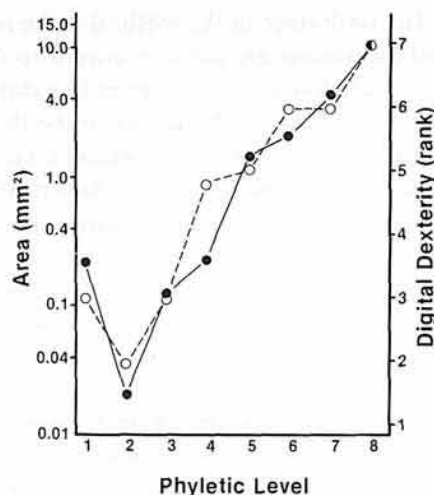


Fig. 5. Changes in the area of the pyramidal tract (filled circles) and digital dexterity (open circles) as a function of phyletic level. Note that increases in the area of the tract parallel improvements in digital dexterity throughout their range.

tained without reversals, the high correlations are determined almost entirely by only two points. Thus, despite the high correlations with dexterity (fig. 4b-d), the early changes in these three parameters do not help to account for the further improvements in dexterity that occurred at higher grades. It is also possible that the limited variation (and, therefore, the limited explanatory potential) in these parameters may be due to their representation here by an ordinal scale. We will return to this point in the discussion.

Discussion

General Considerations

Before turning to a discussion of the results, it is necessary first to note some important limitations on the data and meth-

ods. The limitation in the method is the result of the always arguable relationship of anatomical characters in a series of extant animals to the same characters in the theorized sequence of extinct animals making up mankind's ancestral lineage. This problem is unavoidable and the conclusions must be weakened by readers according to their own analysis of homogenetic versus homoplastic contributions to the structures used here as data. For example, the pyramidal tract has expanded in several Orders of mammals *after* their divergence from ancestral Insectivores and, therefore, this expansion among mammals is *not* homogenetic [Heffner and Masterton, 1975]. Nevertheless, a significant expansion of the tract does *not* seem to have occurred independently within the families of Primates but has occurred among the families and further, this expansion is correlated with propinquity with mankind. Thus, this character seems to be homogenetic in Primates even though it is homoplastic in mammals. Each of the characters discussed here can be subjected to the same kind of analysis when more data become available. At present, we can rely only on correlations (and partial correlations) to eliminate those factors which are probably not related to dexterity, while maintaining a continuous skepticism regarding the factors that *are* reliably correlated.

Beyond the question of deriving evolutionary conclusions from comparative data, there are at least three limitations of the data themselves: The first lies in the coarseness of some of the measurements. For example, subtle but important gradations in digital dexterity might be entirely missed by the present scale. Certainly, an

interval scale devised from careful behavioral tests would eliminate many inaccuracies of approximation based either on anatomy or natural observations alone. Similarly, quantitative measures of the number and the percentage of corticospinal terminals reaching each segment of the spinal cord and each lamina of the spinal cord gray matter might reveal correlations with digital dexterity or phyletic level that may be obscured by the ordinal scales used here. Such precise measures are not presently available, yet their absence might well account for some of the present findings, such as the apparent asymptote of the termination parameters beginning at the level of New World monkeys in figure 3.

Second, information is still missing on certain important species. Very few anatomical descriptions are available for the pyramidal tracts of Insectivora and Prosimiae. These animals represent important grades in man's early ancestry and merit much closer examination. Insectivores in particular, represented here only by hedgehogs, obviously need broader representation. In addition, data on the corticospinal parameters are not complete for all species in the sample and, undoubtedly, the statistical analysis would be clearer if this were not so.

Finally, a methodological difficulty inherent in any correlational analysis is in the determination of cause and effect from statistical correlations. We have attempted to avoid misleading conclusions in several ways. For example, we have applied time constraints – that is, we have considered as possible substrates for dexterity only those anatomical features which showed increases at or before the appearance of im-

provements in digital dexterity. In addition, we have selected variables for study which have previously been implicated in digital dexterity by electrophysiology and ablation experiments. However, we have not attempted to delineate other neurological developments outside the corticospinal system which probably also contributed to the evolution of man's special dexterity. Such developments may have included changes in forebrain, midbrain, and hindbrain motor structures; in somatosensory structures contributing to essential feedback mechanisms; in visual structures (since eye-hand coordination may be important for fine manipulative movements); and in overall brain, frontal cortex, or motor cortex size. The contribution of these aspects of the central nervous system in relation to the evolution of the corticospinal tract and digital dexterity obviously merit examination in their own right.

Despite these limitations and the weakening of the evolutionary implications that they require, several tentative conclusions seem warranted.

Corticospinal Features Important in the Evolution of Digital Dexterity

In general, the present results suggest that the animals thought to be in man's ancestral lineage were probably *not* unique among mammals with respect to the neural control of dexterity. The reason for this conclusion is that the same morphological parameters which were found to be important correlates of digital dexterity among a sample of mammals are also those which are important for the subsample of animals approximating a sequence of mankind's ancestors [cf. *Heffner and*

Masterton, 1975]. Both sets of results suggest an increasing influence of cerebral neocortex over digital movement.

The directness by which cortical neurons can influence motor neurons in the spinal cord seems to depend upon their functional distance from those neurons. One way of reducing this functional distance is to deliver impulses via larger, faster-conducting axons. However, based on data presently available we can only conclude that in man's lineage the increase in the size of corticospinal axons themselves was not beyond that expected from the concurrent increase in overall body size. In other words, conduction became faster in larger bodies (possibly compensating for the increased distance from cortex to motor neurons) but apparently this change was *not* related specifically to digital dexterity.

A second way of reducing the functional distance from cortex to motor neurons is to bypass intervening synapses. This trend does seem to be one that occurred in man's lineage. Additional corticospinal fibers appear which do not synapse in the sensory and internuncial regions of the spinal cord gray but which synapse instead in the vicinity of, or directly on, the motor neuron somata themselves. This trend is illustrated by the correlation between phyletic rank and ventralmost lamina receiving corticospinal terminals. However, due to the concurrent increase in body size in the human lineage and the small number of species on which data are available, this trend becomes less reliable when the influence of body weight is removed mathematically (table IV). (It is appropriate to note here that there has been no noted trend toward fewer terminals in

the more dorsal regions of the spinal gray matter.)

Another change implied in the present data is an early extension of cortical influence beyond the cervical enlargement of the spinal cord to include motor neurons innervating the hindlimbs and tail as well. This trend is documented by the high correlations between phyletic rank, the extension of the corticospinal tract down the spinal cord, and digital dexterity. The extension of cortical influence beyond the cervical level of the spinal cord seems to have had an important, albeit indirect effect on digital dexterity. At the present time we can only suggest that it may have allowed the movements of the forelimbs and hands to become independent of spinal cord locomotor patterns or axial movements, thus freeing them for more dexterous pursuits.

Finally, overall size of the corticospinal tract stands out as the morphological feature most highly correlated both with phyletic level and with digital dexterity. It is consistently parallel with the successive improvements in digital dexterity and it is independent of body weight. Thus, it can be suggested that as the neocortical influence on digital musculature was becoming more direct, the *quantity* of that influence was becoming greater. Though the qualitative changes seem not to have continued among higher primates (see fig. 3b-d) the quantitative changes continued until the emergence of mankind itself.

Corticospinal Features not Correlated with Digital Dexterity

It is important to note that several characteristics of the corticospinal tract, in addition to those emphasized in the results,

have previously been suggested as important indicators of its anatomical development. These include the number of fibers in the tract [Lassek, 1954] and the size of the largest fibers, or fiber spectrum [e.g. Lassek, 1954; Verhaart, 1970a, b]. One purpose of the present analysis was to determine whether these features were also correlates of the development of digital dexterity. The present results, in agreement with Towe [1973], show that the *number of fibers* in the corticospinal tract is primarily a function of body size and that this parameter is not reliably correlated with either phyletic level or dexterity when the influence of body weight is removed. Similarly, both the *size of the largest fibers* and the *average fiber size* are not correlated with either phyletic level or dexterity after the contribution of body size is removed. Of course, the unreliability of these correlations may be due in part to an inadequate sample size and to the difficulty of counting and measuring the numerous small unmyelinated fibers. However, since these features were also found to be uncorrelated with digital dexterity among a much larger sample of mammals [Heffner and Masterton, 1975], we must continue to suspect, until evidence to the contrary becomes available, that their influence in man's line may also have been minimal.

The Evolution of Digital Dexterity

In summary, it would seem appropriate to suggest a course for the evolution of digital dexterity and the accompanying changes in the corticospinal tract based on the data presently available.

Digital dexterity can be said to have undergone qualitative changes up to the ap-

pearance of Eocene Prosimians with the development of a prehensile hand and a pseudo-opposable thumb. Thereafter, changes in dexterity were more in the nature of refining the skill and increasing the independance of the fingers. The corticospinal tract seems to have changed dramatically and in parallel with these early changes in hand use up to the transition from Insectivores to Primates. Instead of synapses restricted to the sensory region of the spinal gray matter, corticospinal synapses began to appear more ventrally, in the internuncial and in the motor region [as suggested by *Noback and Harting, 1971*].

At the same time, cortical neurons extended their direct influence to include even the most caudal segments of the spinal cord and this trend is not restricted to species with tails or bipedal locomotion. Cortical influence over the hindlimbs possibly enabled the forelimbs to function independently from the hindlimbs. The present data suggest that the hands may have been freed for precise digital manipulation by the time of appearance of Anthropoid Primates in the later Eocene which is in close agreement with the fossil evidence that man was bipedal and manufacturing tools early in his history, well before great advances in brain volume occurred [e.g. *Johansson and White, 1979*].

From the mid-Eocene to the present, the corticospinal tract grew larger among the Anthropoid Primates, possibly consolidating its influence over its new neural targets. Again, this increase in tract size parallels the development of a truly opposable thumb and a precision grip among Anthropoids.

Thus, among the early mammals in

mankind's lineage (ground-dwelling and tree-dwelling Insectivores in the early Paleocene), the corticospinal tract was increasing its influence over the somatic musculature by penetrating further down the spinal cord and synapsing nearer to the somata of motor neurons. By the first appearance of Prosimians, the modal synaptic region had shifted ventrally in the spinal gray where it remains in Primates today. At the same time, there was a gradual but continuous trend toward increased size of the corticospinal tract. Together, these trends imply that the corticospinal tract was both broadening its influence over neurons in the spinal cord and making that influence stronger by synapsing directly on motor neuron cell bodies throughout the spinal cord, particularly those cells influencing digital musculature.

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